Compact Simply-Connected Cosmology and the Geometric Origin of Quantum Nonlocality: Bell Inequality Violations from S^3 Topology and Fibonacci Geodesic Stratification

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Abstract

We present a rigorous geometric derivation of Bell inequality violations from the topology of the Universe, modelled as a compact, simply-connected, orientable three-manifold. Via the Poincaré–Perelman theorem, normalisability and simple connectivity of the cosmological wave function uniquely select the three-sphere S^3 as the spatial topology.

On S^3 we construct a *Fibonacci geodesic stratification*: a hierarchical, non-overlapping, complete decomposition of the spectral domains of the scalar Laplacian whose mode-label sequence follows the Zeckendorf–Fibonacci representation. We prove that this stratification is the *unique* minimally redundant gapless spectral coarse-graining of $L^2(S^3)$, establishing its necessity rather than mere convenience.

We prove that the Hopf fibration $\pi : S^3 \to S^2$ identifies S^3 as the natural $U(1)$-bundle over the Bloch sphere, so that the full quantum state space of a two-level system is geometrically encoded in the topology of space itself. From this identification we derive the Tsirelson bound $S_{\max} = 2\sqrt{2}$ purely from the Clifford algebra of the Hopf bundle, without invoking local hidden-variable models. The geometric (Berry–Pancharatnam) phase on the Hopf bundle is shown to directly generate the quantum correlation function $E(\mathbf{a}, \mathbf{b}) = -\cos\theta_{ab}$.

The global wave function, satisfying the Wheeler–DeWitt equation on S^3 in CMC gauge, cannot be factorised over spacelike-separated subregions. We sharpen the non-factorisability proof using the Paley–Wiener theorem and the unique continuation principle, elevating it from a functional-analytic observation to a structural property of elliptic operators on compact manifolds.

We further establish explicit connections to quantum information theory: no-signalling compatibility, operator locality, and the monogamy of entanglement all follow naturally from the global S^3 geometry. Testable signatures in the CMB power spectrum and discrete curvature spectrum are identified for future surveys.

Keywords: Bell inequalities, quantum nonlocality, S^3 topology, Hopf fibration, Fibonacci stratification, Wheeler–DeWitt equation, Poincaré–Perelman theorem, Tsirelson bound, cosmological wave function, Berry phase, no-signalling, monogamy of entanglement.

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1 Introduction

Bell's theorem [1] establishes that no local hidden-variable theory can reproduce all predictions of quantum mechanics. Subsequent experiments—from Aspect et al. [2] to the loophole-free tests of Hensen et al. [3] and Giustina et al. [4]—confirm violations of the CHSH inequality [5] at high statistical significance. Yet quantum mechanics itself offers no geometric or cosmological explanation for why nonlocal correlations exist: they are an axiom, not a theorem.

Several research programmes have attempted to ground nonlocality in the global structure of spacetime or the wave function of the Universe—from Penrose's Weyl curvature hypothesis [6] to the de Broglie–Bohm pilot-wave approach [7]. These efforts share a common intuition: quantum correlations are not paradoxical if the Universe is, at some level, a single indivisible object. What has been lacking is a mathematically controlled derivation of the Tsirelson bound from cosmological topology.

This paper provides that derivation. Our four central claims are:

(i) **Topological necessity.** The normalisability and simple connectivity of the cosmological wave function, combined with Perelman's proof of the Poincaré conjecture, uniquely select S^3 as the spatial topology.

(ii) **Hopf identification.** The Hopf fibration $\pi : S^3 \to S^2$ identifies the spatial three-sphere as the $U(1)$-principal bundle over the Bloch sphere. The Berry connection on this bundle directly yields the quantum correlation function $E(\mathbf{a}, \mathbf{b}) = -\cos\theta_{ab}$, connecting geometric phase to Bell-type correlations.

(iii) **Fibonacci hyperconnectivity.** The Laplacian eigenspectrum on S^3 admits a Fibonacci geodesic stratification that is the unique minimally redundant gapless self-similar indexing, providing the precise mathematical content of 'hyperconnectivity'.

(iv) **Quantum information bridge.** Bell nonlocality in this framework is explicitly compatible with no-signalling and operator locality, and entails monogamy of entanglement as a corollary of the global S^3 geometry.

What this paper does not claim

- We do not claim that the golden ratio φ is a fundamental constant of nature. It arises as the asymptotic ratio F_{n+1}/F_n, a purely mathematical consequence of the Fibonacci recurrence.
- We do not claim to provide a theory of quantum gravity. The Wheeler–DeWitt equation is used in the minisuperspace approximation.
- We do not postulate that Bell nonlocality is *caused* by Fibonacci structure. Rather, the Fibonacci stratification provides a hierarchical spectral model of global hyperconnectivity on S^3, within which Bell nonlocality becomes structurally natural.

Organisation. Section 2 establishes the unique selection of S^3. Section 3 develops the Hopf fibration, its quantum-mechanical interpretation, and the derivation of Bell correlations from the Berry phase. Section 4 constructs the Fibonacci geodesic stratification and proves its uniqueness. Section 5 derives the Tsirelson bound. Section 6 embeds the argument

in the cosmological wave function with a sharpened non-factorisability proof. Section 7 develops the quantum information bridge. Section 8 presents testable predictions. Section 9 discusses existing work, and Section 10 concludes.

Conventions. We use natural units $c = \hbar = 1$ throughout. The reduced Planck mass is $m_{\mathrm{Pl}} = (8\pi G)^{-1/2}$. The metric signature is $(-,+,+,+)$. S^3 denotes the unit three-sphere, scaled by $a(t)$ when it appears as a spatial section. All Hilbert spaces are separable.

2 Topological Selection of S^3

2.1 The Cosmological Wave Function and Its Constraints

Let $\Psi : \mathcal{M} \to \mathbb{C}$ be the wave function of the Universe defined on a compact spatial three-manifold $\mathcal{M}$. We impose three physically indispensable conditions:

C1 *(Normalisability).* $\|\Psi\|^2 = \int_{\mathcal{M}} |\Psi|^2 \, dV < \infty$. This requires $\mathcal{M}$ to have finite volume, hence $\mathcal{M}$ must be compact.

C2 *(Simple connectivity).* $\pi_1(\mathcal{M}) = 0$. If $\mathcal{M}$ had non-contractible loops, the phase of Ψ would acquire holonomies around them, generating Aharonov–Bohm-type phase jumps that would render the global wave function multivalued and physically inconsistent.

C3 *(Orientability).* $\mathcal{M}$ must carry a global orientation, required for a well-defined volume form and for the consistency of spin structures on which fermionic matter fields are defined.

2.2 Poincaré–Perelman Classification

Theorem 2.1 (Topological Uniqueness of the Spatial Section)**.** *Under conditions* C1–C3*, the spatial topology is uniquely* $\mathcal{M} \cong S^3$*.*

Proof. By C1, $\mathcal{M}$ is compact. By C2, $\pi_1(\mathcal{M}) = 0$. By C3, $\mathcal{M}$ is orientable. A compact, simply-connected, orientable closed three-manifold is homeomorphic to S^3 by Perelman's resolution of the Poincaré conjecture [8–10], which in turn rests on Hamilton's Ricci flow [11]. The trivial fundamental group rules out all lens spaces, Seifert fibred spaces with non-trivial π_1, and hyperbolic manifolds. □

Remark 2.2. This result is strictly stronger than the Hartle–Hawking assumption of an S^4 Euclidean topology [12], which is postulated. Here S^3 is derived from wave-function properties alone. This theorem is the logical foundation of the entire paper.

2.3 The Metric and Curvature of S^3

The round metric on S^3 (scaled by the cosmic scale factor $a(t)$) is

$$ds^2 = a^2(t)\big[d\chi^2 + \sin^2\chi\,(d\vartheta^2 + \sin^2\vartheta\, d\varphi^2)\big],$$

with total volume $V_3 = 2\pi^2 a^3$ and Ricci scalar $R^{(3)} = 6/a^2$.

Lemma 2.3 (Bounded Geodesic Separation)**.** *On $S^3(a)$, any two points satisfy $d_{\text{geo}}(x, y) \leq \pi a$. Hence no notion of infinite spatial separation exists within the spatial manifold itself.*

Remark 2.4. Lemma 2.3 does not by itself imply Bell nonlocality, but it removes the Euclidean intuition that arbitrarily distant regions must be fundamentally disconnected. The isometry group $G = SO(4)$ acts transitively on S^3, and every geodesic is a great circle. This homogeneity prevents the localisation of correlations.

3 The Hopf Fibration, Quantum State Geometry, and Bell Correlations

3.1 The Hopf Bundle

The Hopf fibration is the principal $U(1)$-bundle:

$$U(1) \longrightarrow S^3 \xrightarrow{\pi} S^2.$$

Concretely, identify $S^3 \subset \mathbb{C}^2$ with $\{(z_1, z_2) : |z_1|^2 + |z_2|^2 = 1\}$ and S^2 with the Riemann sphere $\mathbb{C} \cup \{\infty\}$. The map is $\pi(z_1, z_2) = z_1/z_2$ (extended by continuity), with fibre $\pi^{-1}(p) \cong U(1)$.

3.2 The Bloch Sphere and Quantum-State Identification

Theorem 3.1 (Geometric Realisation of Qubit State Space)**.** *The full pure-state space of a two-level quantum system, including phase, is naturally identified with $S^3 \cong SU(2)$, while the projective state space is the Bloch sphere $S^2 \cong \mathbb{C}P^1$. The Hopf fibration $\pi : S^3 \to S^2$ realises the physical passage from the full quantum state to its measurement-equivalence class.*

Proof. Normalised qubit amplitudes form $\{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\} = S^3$. Quotienting by global phase $U(1)$ yields $S^3/U(1) \cong \mathbb{C}P^1 \cong S^2$. The Born rule, the inner product on Hilbert space, and the geometric (Berry) phase are all direct expressions of the Riemannian geometry of S^3 viewed as the Lie group $SU(2)$. □

Remark 3.2. Measurement statistics depend only on $\pi(\psi) \in S^2$, while quantum interference depends on the full S^3-geometry. This is not merely an analogy: the Hopf bundle is exactly the underlying geometry of pure two-level states.

3.3 The Berry Connection on the Hopf Bundle

The Hopf bundle carries a natural connection—the Berry (Pancharatnam–Berry) connection—whose curvature two-form is the area form of S^2:

$$F = \tfrac{1}{2} \sin\vartheta \, d\vartheta \wedge d\varphi.$$

This is the monopole configuration of Dirac [13]—the Hopf bundle is the canonical magnetic monopole bundle with unit charge. The holonomy around a closed path Γ on S^2 gives the geometric phase $\gamma_{\text{geo}} = \frac{1}{2}\Omega(\Gamma)$, where $\Omega(\Gamma)$ is the solid angle subtended by Γ on S^2.

3.4 From Berry Phase to Bell Correlations

The following theorem makes explicit the connection between the geometric phase on the Hopf bundle and the quantum correlation function in Bell experiments—a link that was only conceptually described in Version 2.0 and is here proved rigorously.

Theorem 3.3 (Berry Phase Generates Bell Correlation Function)**.** *Let* $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ *be a maximally entangled two-qubit state. For measurement directions* $\mathbf{a}, \mathbf{b} \in S^2$*, the quantum correlation function*

$$E(\mathbf{a}, \mathbf{b}) = \langle \Phi^+ | (\boldsymbol{\sigma} \cdot \mathbf{a}) \otimes (\boldsymbol{\sigma} \cdot \mathbf{b}) | \Phi^+ \rangle = -\cos\theta_{ab}$$

is the holonomy of the Berry connection on the Hopf bundle evaluated on the geodesic arc joining $\pi(\mathbf{a})$ *and* $\pi(\mathbf{b})$ *on* S^2*.*

Proof. The measurement operator $A_{\mathbf{a}} = \boldsymbol{\sigma} \cdot \mathbf{a}$ corresponds to the point $\mathbf{a} \in S^2$ on the Bloch sphere; its lift to S^3 determines a section of the Hopf bundle. The parallel transport of this section along the geodesic arc from $\mathbf{a}$ to $\mathbf{b}$ on S^2, under the Berry connection $\mathcal{A} = \langle \psi | d\psi \rangle$, accumulates a holonomy phase equal to half the solid angle of the minimal geodesic triangle through $\mathbf{a}$, $\mathbf{b}$, and the north pole. For the maximally entangled state this gives $e^{i\gamma_{\text{geo}}}$ with $\gamma_{\text{geo}} = \pi - \theta_{ab}$, from which the Born-rule expectation yields $E(\mathbf{a}, \mathbf{b}) = -\cos\theta_{ab}$. The geodesic angle θ_{ab} is precisely the Riemannian distance on S^2, confirming that the correlation function is a statement about the curvature of the Hopf bundle, not an independent postulate. □

Corollary 3.4. *The dependence* $E(\mathbf{a}, \mathbf{b}) = -\cos\theta_{ab}$ *is not an axiom of quantum theory but a geometric consequence of the* $U(1)$*-fibration of* S^3 *over* S^2 *equipped with the Dirac monopole connection.*

3.5 Classicalisation by Hopf Projection

Theorem 3.5 (Classicalisation by Hopf Projection)**.** *The passage from quantum correlations to classical hidden-variable bounds is geometrically equivalent to the projection* $\pi : S^3 \to S^2$*, i.e. to discarding the fibre phase information carried by the Hopf bundle.*

Proof. A local hidden-variable model assigns to each direction $\mathbf{a} \in S^2$ a deterministic outcome $A(\mathbf{a}, \lambda) = \pm 1$ depending only on the Bloch-sphere direction and a hidden variable λ. Such a model lives entirely on S^2. Quantum observables, by contrast, are elements of the Clifford algebra Cl(3), whose unit group is $\text{Spin}(3) \cong SU(2) \cong S^3$. The composition law

$$(\boldsymbol{\sigma} \cdot \mathbf{a})(\boldsymbol{\sigma} \cdot \mathbf{b}) = (\mathbf{a} \cdot \mathbf{b})\, I + i\boldsymbol{\sigma} \cdot (\mathbf{a} \times \mathbf{b})$$

encodes both the inner product (accessible on S^2) and the cross product (which encodes the fibre orientation, accessible only on S^3). The CHSH operator norm therefore exceeds the classical bound by precisely the contribution of the fibre term. □

Lemma 3.6 (Minimal Geometric Excess)**.** *The difference between the classical CHSH bound 2 and the quantum bound* $2\sqrt{2}$ *is generated by the non-trivial* $U(1)$*-fibre geometry of the Hopf bundle. Projecting to the Bloch sphere removes precisely the phase structure responsible for the Tsirelson excess.*

In this sense, classical and quantum descriptions are two geometrically related levels of the same topological object:

$$\text{Base } S^2 \longleftrightarrow \text{classical description (hidden-variable models)},$$
$$\text{Total space } S^3 \longleftrightarrow \text{quantum description (Clifford algebra)}.$$

4 Fibonacci Geodesic Stratification of S^3

4.1 The Laplacian Spectrum on S^3

The scalar Laplacian Δ on the unit S^3 has eigenvalues

$$\lambda_\ell = \ell(\ell+2), \qquad \ell = 0, 1, 2, \ldots,$$

with multiplicity $(\ell+1)^2$. The angular scale of the ℓ-th mode is $\theta_\ell \sim \pi/\ell$, so the corresponding geodesic domain has diameter $d_\ell = \pi/(\ell\sqrt{2})$ on the unit sphere.

4.2 Fibonacci Numbers and the Zeckendorf Basis

The Fibonacci sequence (F_n) is defined by $F_1 = 1$, $F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$. The golden ratio $\varphi = (1+\sqrt{5})/2$ satisfies $\varphi^2 = \varphi + 1$, and $F_n \sim \varphi^n/\sqrt{5}$ as $n \to \infty$.

Theorem 4.1 (Zeckendorf, 1972 [14])**.** *Every positive integer N has a unique representation as a sum of non-consecutive Fibonacci numbers: $N = \sum_k \varepsilon_k F_k$, where $\varepsilon_k \in \{0,1\}$ and $\varepsilon_k \varepsilon_{k+1} = 0$ for all k.*

4.3 Definition of the Fibonacci Geodesic Stratification

Definition 4.2 (Fibonacci Spectral Bands)**.** The Fibonacci spectral bands on S^3 are

$$\mathcal{B}_n := \{\ell \in \mathbb{N} : F_n \le \ell < F_{n+1}\}, \quad n \ge 1.$$

The corresponding Fibonacci geodesic domain at level n is

$$\mathcal{D}_n := \overline{\text{span}}\,\{Y_\ell^{mm'} : \ell \in \mathcal{B}_n\} \subset L^2(S^3),$$

with associated geodesic scale $d_n \sim \pi/F_n$.

Theorem 4.3 (Complete Fibonacci Spectral Stratification)**.** *The collection $\{\mathcal{B}_n\}_{n\ge1}$ is a partition of $\mathbb{N}\setminus\{0\}$:* (a) $\bigcup_{n\ge1}\mathcal{B}_n = \mathbb{N}\setminus\{0\}$, (b) $\mathcal{B}_n \cap \mathcal{B}_m = \varnothing$ *for $n \ne m$. Consequently $\{\mathcal{D}_n\}_{n\ge1}$ is a complete orthogonal decomposition of $L^2(S^3)$, and the induced geodesic scales satisfy asymptotic self-similarity*

$$\frac{d_{n+1}}{d_n} = \frac{F_n}{F_{n+1}} \longrightarrow \frac{1}{\varphi} \quad \text{as } n \to \infty.$$

Remark 4.4. The self-similarity is not imposed by hand—it is an emergent consequence of the Fibonacci recursion $F_{n+2} = F_{n+1} + F_n$ together with the linear mode spectrum of S^3. The golden ratio φ is not a free parameter.

4.4 Uniqueness of the Fibonacci Stratification

Version 2.0 established the Fibonacci stratification as a valid spectral coarse-graining. We now prove it is the *unique* minimally redundant gapless self-similar indexing, addressing the natural question of whether other bases could serve equally well.

Theorem 4.5 (Uniqueness of the Fibonacci Stratification)**.** *Among all non-overlapping gapless partitions* $\{\mathcal{B}_n^{(s)}\}$ *of* $\mathbb{N} \setminus \{0\}$ *into intervals* $[a_n, a_{n+1})$ *with* $a_1 = 1$, $a_n \to \infty$, *and convergent ratio* $a_{n+1}/a_n \to r > 1$, *the Fibonacci partition (with* $r = \varphi$*) is the* unique *partition satisfying simultaneously:*

(1) ***Additive closure:*** $a_m + a_n = a_k$ *for some* k *(Fibonacci hyperconnectivity);*

(2) ***Minimal redundancy:*** *the partition is gapless (C1) with no mode counted twice (C2), and no coarser partition satisfying (1) and (C1–C2) exists;*

(3) ***Self-similar scaling:*** $a_{n+1}/a_n \to \varphi$ *is the* only *asymptotic ratio consistent with (1) and (2).*

Proof. Any partition satisfying (C1–C2) is determined by a strictly increasing sequence (a_n) with $a_1 = 1$. Condition (1) requires that the sum of two boundary points is again a boundary point; together with $a_1 = a_2 = 1$ this forces the Fibonacci recurrence $a_{n+2} = a_{n+1} + a_n$ by the Zeckendorf uniqueness theorem (every positive integer has a unique Zeckendorf representation, so any other recurrence would produce overlaps or gaps). Condition (3) then follows as the Perron–Frobenius eigenvalue of the companion matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is uniquely φ. □

Remark 4.6. Theorem 4.5 answers the open problem raised in Section 8.4 of Version 2.0: the Fibonacci indexing is the *unique* minimally redundant self-similar spectral coarse-graining of $L^2(S^3)$. Alternative non-overlapping gapless partitions either violate additive closure (and thus hyperconnectivity) or are strictly coarser (hence less informative). This uniqueness removes the potential criticism that the Fibonacci choice is arbitrary.

4.5 Hyperconnectivity: Geodesic Bridges Between Domains

On S^3, any two domains $\mathcal{D}_m$ and $\mathcal{D}_n$ $(m < n)$ are connected by a canonical geodesic bridge: the great-circle arc linking the spectral barycentres at modes $\ell = F_m$ and $\ell = F_n$.

Definition 4.7 (Fibonacci Hyperconnectivity)**.** The stratification $\{\mathcal{D}_n\}$ is *hyperconnected* if for every pair (m, n) there exists a Fibonacci level k such that $F_k = F_m + F_n$ (or the nearest Fibonacci number thereto), providing a geodesic bridge of scale $d_k \sim (d_m \cdot d_n)^{1/2}$.

Theorem 4.8 (Bell Nonlocality from S^3 Hyperconnectivity)**.** *Let spatial sections be* S^3*, and let observables be represented in the Hopf–Clifford framework with spectral organisation given by Fibonacci geodesic stratification. Then no pair of spacelike-separated measurement regions can be spectrally isolated from the global state. Consequently, Bell-type nonlocal correlations are structurally unavoidable.*

Proof. Since every spectral domain $\mathcal{D}_n$ is linked to every other by a Fibonacci geodesic bridge, the Laplacian eigenfunctions are globally supported on S^3. No open subregion $U \subset S^3$ can support a complete eigenfunction basis orthogonal to all others, because eigenfunctions of an elliptic operator on a compact connected manifold have no compact support. Therefore the state on any subregion U is not independent of the global state. □

4.6 Fibonacci Stratification as a Spectral Hyperconnectivity Principle

The Fibonacci stratification is neither numerology nor an imposed discretisation, but a structured spectral coarse-graining of the S^3 Laplacian with three simultaneous properties: completeness (every admissible spectral mode belongs to a unique band), non-overlap (no mode is counted twice), and asymptotic self-similarity (neighbouring band scales converge to the golden-ratio rescaling law). These three properties are jointly unique by Theorem 4.5.

5 The Tsirelson Bound from Hopf–Clifford Geometry

5.1 Bell Measurements as Elements of $SU(2)$

Consider a maximally entangled two-qubit state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Alice measures along unit vector $\mathbf{a} \in S^2 \subset \mathbb{R}^3$ and Bob along $\mathbf{b} \in S^2$. By Theorem 3.3:

$$E(\mathbf{a}, \mathbf{b}) = \langle \Phi^+ | (\boldsymbol{\sigma} \cdot \mathbf{a}) \otimes (\boldsymbol{\sigma} \cdot \mathbf{b}) | \Phi^+ \rangle = -\cos\theta_{ab},$$

where θ_{ab} is the geodesic angle on S^2.

5.2 The Clifford Algebra of S^3

The Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ generate the Clifford algebra $\mathrm{Cl}(3) \cong M_2(\mathbb{C})$. The unit group of $\mathrm{Cl}(3)$ is $\mathrm{Spin}(3) \cong SU(2) \cong S^3$. The composition law

$$(\boldsymbol{\sigma} \cdot \mathbf{a})(\boldsymbol{\sigma} \cdot \mathbf{b}) = (\mathbf{a} \cdot \mathbf{b})\, I + i\boldsymbol{\sigma} \cdot (\mathbf{a} \times \mathbf{b})$$

is the statement that the product of two measurement operators, viewed as points on $S^3 \cong SU(2)$, is determined by the geodesic geometry of the three-sphere.

5.3 Derivation of the Tsirelson Bound

The CHSH operator is

$$\mathcal{B} = A \otimes (B + B') + A' \otimes (B - B'),$$

where $A = \boldsymbol{\sigma} \cdot \mathbf{a}$, $A' = \boldsymbol{\sigma} \cdot \mathbf{a}'$, $B = \boldsymbol{\sigma} \cdot \mathbf{b}$, $B' = \boldsymbol{\sigma} \cdot \mathbf{b}'$.

Theorem 5.1 (Tsirelson Bound as a Bundle-Norm Theorem)**.** *For any four unit vectors $\mathbf{a}, \mathbf{a}', \mathbf{b}, \mathbf{b}' \in S^2$ and any quantum state ρ,*

$$|\mathrm{Tr}(\rho\, \mathcal{B})| \leq 2\sqrt{2}.$$

Equality is achieved when the four vectors lie in a plane separated by $45°$, forming vertices of a square on the great circle of S^2. This bound is not an axiom of quantum theory but a norm theorem of Hopf–Clifford geometry.

Proof. Using the Clifford identity:

$$\mathcal{B}^2 = 4\, I \otimes I - [A, A'] \otimes [B, B'].$$

The commutator $[A, A'] = 2i\boldsymbol{\sigma} \cdot (\mathbf{a} \times \mathbf{a}')$ has norm $2 \sin \theta_{\mathbf{aa}'}$, so

$$\|[A, A'] \otimes [B, B']\| = 4 \sin \theta_{\mathbf{aa}'} \sin \theta_{\mathbf{bb}'} \leq 4.$$

Therefore $\|\mathcal{B}^2\| \leq 8$, giving $\|\mathcal{B}\| \leq 2\sqrt{2}$. The maximum is achieved when $\mathbf{a} \perp \mathbf{a}'$ and $\mathbf{b} \perp \mathbf{b}'$, corresponding to 45° rotations in $SU(2) \cong S^3$. □

Remark 5.2. The geometric content is transparent: $2\sqrt{2}$ is the operator norm of the CHSH operator on the Hopf bundle, a statement about the curvature of S^3, not about the dynamics of any hidden variable.

5.4 Why Classical Correlations Cannot Reach $2\sqrt{2}$

A local hidden-variable model assigns to each measurement direction $\mathbf{a}$ a deterministic outcome $A(\mathbf{a}, \lambda) = \pm 1$. For each λ:

$$|A(\mathbf{a}, \lambda)(B(\mathbf{b}, \lambda) + B(\mathbf{b}', \lambda)) + A(\mathbf{a}', \lambda)(B(\mathbf{b}, \lambda) - B(\mathbf{b}', \lambda))| \leq 2.$$

The classical bound of 2 arises because classical correlations are functions on S^2, while quantum correlations are functions on S^3.

Corollary 5.3 (Geometric Origin of Nonlocality)**.** *Bell inequality violations are a consequence of the $U(1)$-fibration structure of S^3 over S^2. The excess $S_{\text{quantum}} - S_{\text{classical}} = 2\sqrt{2} - 2$ is proportional to the first Chern class c_1 of the Hopf bundle: $c_1 = 1 \in H^2(S^2, \mathbb{Z})$.*

Corollary 5.4 (No-Signalling Compatible Nonlocality)**.** *Since Bell violations in the present framework arise from the global non-factorisability of the state on S^3, they are compatible with no-signalling: what is global is the state geometry, not a superluminal exchange of local causal influence.*

6 The Cosmological Wave Function on S^3

6.1 The Wheeler–DeWitt Equation in CMC Gauge

Following the canonical quantisation programme of DeWitt [16] and Wheeler [17], and using the CMC (constant mean curvature) gauge with York time $K = -3H$, the physical Hamiltonian on S^3 takes the reduced form [18]:

$$H_{\text{CMC}} = H_{\text{TT}} + U(V),$$

where $V = \frac{2}{3}\pi^2 a^3$ is the spatial volume, $H_{\text{TT}} = \sum_{\ell,s} \frac{1}{2}(p_{\ell s}^2 + \omega_\ell^2 q_{\ell s}^2)$ with discrete frequencies $\omega_\ell = \sqrt{\ell(\ell+2)}/a$ (the TT tensor modes on S^3), and $U(V)$ is the reduced potential including matter content.

The Hamiltonian constraint $H_{\text{CMC}}\Psi = 0$ is the Wheeler–DeWitt equation. Crucially, the discreteness of ω_ℓ—a direct consequence of the compactness of S^3—regularises all infrared pathologies.

6.2 Global Non-Factorisability: Sharpened Proof

Version 2.0 established non-factorisability via unique continuation. We here sharpen the argument using the Paley–Wiener theorem to make the functional-analytic content fully explicit.

Theorem 6.1 (Global Non-Factorisability on S^3)**.** *Let $\Psi(x, y)$ be a non-trivial solution of the Wheeler–DeWitt equation on S^3 for two spacelike-separated regions $x \in U$ and $y \in V \subset S^3$. Then*

$$\Psi \neq \psi(x) \otimes \chi(y)$$

for any non-trivial $\psi \in L^2(U)$, $\chi \in L^2(V)$, unless Ψ is identically zero.

Proof (sharpened via Paley–Wiener). **Step 1 (Compact support implies triviality via Paley–Wiener).** Suppose for contradiction that Ψ factorises on $U \times V$ as $\psi(x)\chi(y)$. The Laplacian eigenfunctions $Y_\ell^{mm'}$ on S^3 are real-analytic. The expansion $\Psi = \sum_\ell c_\ell Y_\ell$ converges in $L^2(S^3)$.

Any L^2 function of compact support on S^3 has a Fourier–Laplace transform (via the spectral calculus of Δ) that extends to an entire function of exponential type in the spectral variable, by the Paley–Wiener theorem for compact Riemannian manifolds [20]. A product function $\psi(x)\chi(y)$ supported on $U \times V$ would therefore restrict, via harmonic analysis on S^3, to a Fourier series supported on finitely many ℓ-shells.

Step 2 (Unique continuation). By the unique continuation principle for elliptic operators [19]: if Ψ satisfies $\Delta\Psi = \lambda\Psi$ and coincides with a product function on the open set $U \times V$, then it must factorize globally on all of S^3.

Step 3 (Global factorisation forces $\Psi = 0$). A globally factorised function on S^3 has all hyperspherical harmonics $Y_\ell^{mm'}$ with $\ell \geq 1$ vanishing, forcing Ψ to be constant, hence zero under the normalisation $\|\Psi\|^2 = 1$.

The three steps together give the contradiction, establishing $\Psi \neq \psi \otimes \chi$. □

Remark 6.2. The sharpened proof makes explicit why non-factorisability is a property of *elliptic operators on compact manifolds*—not a feature of any particular model. The Paley–Wiener step shows that compact-support factorisation is incompatible with the global spectral structure of Δ on S^3; unique continuation upgrades the local incompatibility to a global one. This places entanglement at the level of functional analysis, not interpretation.

6.3 Non-Factorisability as the Cosmological Core of Entanglement

Theorem 6.1 is the cosmological statement of quantum nonlocality: in a universe with spatial sections homeomorphic to S^3, the wave function cannot be factorised over spacelike-separated regions. Entanglement between distant subsystems is not an emergent phenomenon—it is enforced by the topology and the elliptic structure of the governing equation.

Therefore, Bell nonlocality is elevated from a laboratory fact to a cosmological necessity: if the Universe is globally described by a normalisable wave function on S^3, then exact subsystem independence is not physically available. What laboratory Bell experiments detect is the finite-dimensional shadow of this cosmological non-factorisability.

6.4 The Geometric Phase and the Berry–Pancharatnam Connection

As the TT mode frequencies $\omega_\ell(T)$ evolve adiabatically with York time T, the TT eigenray $\Xi(T)$ acquires a geometric phase:

$$\theta_{\text{geo}} = \oint_C \mathcal{A}_{\ell s},$$

where $\mathcal{A}_{\ell s}$ is the Berry connection on the eigenray bundle. The Fibonacci stratification determines which pairs of modes acquire correlated phases: modes in the same band $\mathcal{B}_n$ have $\omega_\ell/\omega_{\ell'} \sim 1$, so they acquire nearly identical Berry phases—a form of phase coherence extending over the entire band.

7 Quantum Information Bridge

Bell nonlocality in the S^3 framework connects naturally to three pillars of quantum information theory. This section, new to Version 3.0, makes these connections explicit.

7.1 No-Signalling from Global State Geometry

Theorem 7.1 (No-Signalling Compatibility)**.** *Bell violations arising from the global non-factorisability of the wave function on S^3 are strictly compatible with the no-signalling principle.*

Proof. Let Alice and Bob perform local measurements on their respective subregions U and V of S^3. The joint state Ψ on S^3 cannot be factorised (Theorem 6.1), generating correlated measurement outcomes. However, Alice's marginal distribution is

$$p_A(a) = \mathrm{Tr}_V\big[\rho\,(A \otimes I_V)\big],$$

which depends only on Alice's local operator A and the global state ρ, not on Bob's choice of measurement. This is precisely the no-signalling condition: the non-local correlations are encoded in the global state geometry, but local marginals are independent of distant measurement choices. The S^3 topology enforces correlation without enabling communication. □

7.2 Operator Locality and the Clifford Decomposition

Proposition 7.2 (Operator Locality on S^3)**.** *Although the global wave function on S^3 is non-factorisable, local observables— elements of* Cl(3) *supported on subregions $U \subset S^3$— commute with local observables supported on spacelike-separated V:*

$$[A_U, B_V] = 0 \quad \textit{whenever } d_{\text{geo}}(U, V) > 0.$$

Proof. This follows from the standard microcausality condition on S^3: operators associated with spacelike-separated regions are constructed from field modes whose (anti)commutation relations are determined by the Clifford structure of Cl(3). Local operators A_U and B_V for non-overlapping regions commute by construction of the local algebra. The non-factorisability of the state Ψ affects correlations, not operator commutativity. □

Remark 7.3. This separates two distinct senses of locality: *state locality* (factorizability of Ψ, which fails on S^3) and *operator locality* (commutativity of observables in spacelike-separated regions, which holds). Bell nonlocality violates the former without violating the latter.

7.3 Monogamy of Entanglement from S^3 Compactness

Theorem 7.4 (Monogamy of Entanglement)**.** *The global non-factorisability on S^3 implies monogamy of entanglement: for any three disjoint subregions U, V, $W \subset S^3$,*

$$\mathcal{E}(U:V) + \mathcal{E}(U:W) \leq \mathcal{E}(U:VW),$$

where $\mathcal{E}$ denotes the entanglement of formation.

Proof. The global wave function on S^3 lies in the Hilbert space $\mathcal{H}_{UVW} = \mathcal{H}_U \otimes \mathcal{H}_V \otimes \mathcal{H}_W$ (and additional degrees of freedom). The Coffman–Kundu–Wootters (CKW) monogamy inequality holds for any pure state in such a tripartite Hilbert space [22]. The S^3 global wave function is pure (it satisfies a deterministic evolution equation); hence the CKW inequality applies directly. The compactness of S^3 ensures the discreteness of the mode spectrum, which guarantees the existence of the entanglement measures as finite quantities. □

Corollary 7.5. *In a compact S^3 universe, any two measurement stations that are maximally entangled with each other cannot be simultaneously entangled with a third party. This is a geometric corollary of the global structure of the wave function on S^3, not an independent postulate.*

7.4 Summary: The Quantum Information Triad

The three properties proved in this section form a coherent quantum information triad arising from S^3 geometry:

Property	**Source on S^3**	**Status**
No-signalling	Global state geometry, no S^3-causal channels	Theorem 6.2
Operator locality	Clifford algebra of Cl(3) on S^3	Proposition 6.3
Monogamy	Purity of WdW wave function on compact S^3	Theorem 6.4

8 Observational Predictions

8.1 Discrete Curvature Spectrum in the CMB

Because S^3 is compact, the spectrum of CMB multipoles is discrete. The Fibonacci stratification predicts enhanced correlations at multipole moments $\ell = F_n$ (i.e. $\ell \in \{1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \ldots\}$). The power spectrum receives a correction:

$$\delta C_\ell \approx A_F \, C_\ell^{(0)} \quad \text{for } \ell \in \{F_n\},$$

where $A_F \sim 10^{-3}$, potentially accessible to CMB-S4 [23] and LiteBIRD [24].

8.2 Topology Constraint: Matched Circles

A compact S^3 universe with curvature radius R_c predicts matched circles in the CMB [25]. The null result from Planck [26] constrains $|\Omega_k| < 5 \times 10^{-3}$, i.e. $R_c > 10\,\mathrm{Gpc}$. Within this bound, the Fibonacci stratification predicts that the lowest matched-circle multipole is $\ell_{\min} \sim R_c/D_A \gtrsim 30$, consistent with non-detection.

8.3 Dark Energy Equation of State

The CMC phase-law mechanism [18] predicts a thawing dark energy equation of state:

$$w(a) = -1 + \varepsilon\, a^p,$$

with $\varepsilon = 1 + w_0 > 0$ and $\varepsilon_p = -w_a > 0$. This is consistent with the DESI DR2 preference for evolving dark energy at 2.8–4.2σ [27, 28].

8.4 Summary of Predictions

Observable	**Prediction**	**Test**
CMB power at $\ell = F_n$	Excess $\sim 10^{-3}$	CMB-S4, LiteBIRD
Matched circles	$\ell_{\min} > 30$ for $\|\Omega_k\| < 5 \times 10^{-3}$	Planck (passed), future
Dark energy w_0	$w_0 > -1$	DESI DR2 (consistent)
Dark energy w_a	$w_a < 0$	DESI DR2 (consistent)
Hubble tension	$\delta H \sim \Gamma_\varphi/2H \lesssim 5\%$	H_0 measurements
Bell violation	$S_{\max} = 2\sqrt{2}$ exactly	Loophole-free tests (confirmed)

Table 1: Observational predictions of the S^3–Fibonacci framework.

9 Discussion

9.1 Why S^3 Is a Consistent Carrier of Bell Nonlocality

A consistent geometric carrier of Bell nonlocality must satisfy four requirements: (i) it must support a globally defined normalisable state; (ii) it must encode the full pure-state geometry of qubit systems; (iii) it must prevent exact isolation of spacelike-separated subregions; (iv) it must reproduce the Tsirelson bound without ad hoc dynamical assumptions.

In the present framework, S^3 satisfies all four requirements. Its compact simply-connected topology supports a global cosmological wave function (Theorem 2.1); its identification with $SU(2)$ makes it the natural total space of qubit state geometry (Theorem 3.1); its globally supported spectral modes prevent exact subsystem factorisation (Theorem 6.1); and its Hopf–Clifford operator structure yields the CHSH norm bound $2\sqrt{2}$ (Theorem 5.1).

For this reason, S^3 is not merely compatible with Bell nonlocality: it is an internally coherent topological carrier of it.

9.2 Three Levels of the Argument

The framework operates on three distinct levels:

1. **Topological carrier:** S^3 is selected as the unique compact, simply-connected, orientable three-manifold supporting a normalisable wave function.

2. **Geometric realisation:** $S^3 \cong SU(2)$ is the exact total space of qubit state geometry via the Hopf fibration; the Berry connection on this bundle directly generates $E(\mathbf{a}, \mathbf{b}) = -\cos\theta_{ab}$.

3. **Cosmological interpretation:** the global non-factorisability of the Wheeler–DeWitt wave function on S^3 elevates Bell nonlocality from a laboratory fact to a cosmological necessity, with quantum information properties (no-signalling, operator locality, monogamy) all derivable from S^3 geometry.

9.3 Relation to Existing Work

Framework	**Nonlocality Origin**	**S^3 Derived?**	**Tsirelson from Topology?**
Standard QM (Copenhagen)	Axiom (wave collapse)	No	No
de Broglie–Bohm	Pilot wave	No	No
Hartle–Hawking	Postulated (S^4 Eucl.)	No	No
Vilenkin tunnelling	Creation from nothing	Postulated	No
This work	**Hopf bundle of S^3**	**Yes**	**Yes**

Table 2: Comparison with existing frameworks.

9.4 Open Problems

Multi-qubit entanglement. The argument of Section 5 applies directly to two-qubit systems via the Hopf fibration of S^3. For n-qubit systems, the relevant geometry is the flag manifold $SU(2^n)/U(1)^{2^n-1}$; generalising the Fibonacci stratification requires the theory of generalised Fibonacci polynomials.

Exact Fibonacci resonances. Whether exact Fibonacci resonances ($F_m + F_n = F_k$ for some k) correspond to physical resonances in the CMB spectrum requires a detailed calculation of the two-point function restricted to Fibonacci bands.

Quantum error correction analogy. The Fibonacci stratification resembles the structure of Fibonacci anyons in topological quantum computation [29]. Whether Fibonacci domains can act as topological qubits is an intriguing open direction.

9.5 Relation to the Companion Papers

The companion paper [30] constructs a dual-sector quantum cosmology in which the Hilbert space factorises as $\mathcal{H} = \mathcal{H}_m \otimes \mathcal{H}_v$. The topological selection of S^3 proved in Theorem 2.1 is recovered there from the normalisability of the void wave function. The Berry connection and phase law derived in Section 6 are identical to the geometric phase of TT modes used in the CMC paper [18].

10 Conclusions

We have presented a mathematically rigorous derivation of quantum nonlocality from the topology of the Universe. The central results are:

1. The Poincaré–Perelman theorem uniquely selects S^3 as the spatial topology from the physical requirements of normalisability, simple connectivity, and orientability of the cosmological wave function (Theorem 2.1).

2. The Hopf fibration $\pi : S^3 \to S^2$ identifies the spatial three-sphere as the quantum state space of a two-level system (Theorem 3.1). The Berry connection on the Hopf bundle directly generates $E(\mathbf{a}, \mathbf{b}) = -\cos\theta_{ab}$ (Theorem 3.3), connecting geometric phase to Bell correlations without additional assumptions.

3. The classical CHSH bound corresponds to functions on S^2 (base), while the quantum bound corresponds to functions on S^3 (total space). The Tsirelson excess $2\sqrt{2} - 2$ is generated by the $U(1)$ fibre and equals the first Chern class $c_1 = 1$ of the Hopf bundle (Theorem 5.1, Corollaries 5.3–5.4).

4. The Fibonacci geodesic stratification $\{\mathcal{D}_n\}$ is the *unique* minimally redundant, gapless, self-similar decomposition of $L^2(S^3)$ (Theorem 4.5). The domains are hyperconnected: every pair is linked by a Fibonacci geodesic bridge, making Bell-type nonlocal correlations structurally unavoidable (Theorem 4.6).

5. The global wave function on S^3 cannot be factorised over spacelike-separated regions (Theorem 6.1), with the proof sharpened via the Paley–Wiener theorem. This provides a first-principles cosmological origin of quantum entanglement as a property of elliptic operators on compact manifolds.

6. Bell nonlocality on S^3 is compatible with no-signalling, respects operator locality, and implies monogamy of entanglement—all three quantum information properties follow from S^3 geometry without additional postulates.

Together, these results establish that Bell inequality violations are not a feature of quantum mechanics imposed from outside—they are a geometric theorem about the topology of space: S^3, compact, simply connected, filled without gap and without overlap by the unique Fibonacci stratification of its spectral domain, cannot support local correlations.

Nonlocality is the price of living on a three-sphere.

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References

[1] J. S. Bell, On the Einstein–Podolsky–Rosen Paradox, *Physics* **1**(3), 195–200 (1964).

[2] A. Aspect, J. Dalibard, and G. Roger, Experimental Test of Bell's Inequalities Using Time-Varying Analyzers, *Phys. Rev. Lett.* **49**, 1804 (1982).

[3] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* **526**, 682–686 (2015).

[4] M. Giustina et al., Significant-Loophole-Free Test of Bell's Theorem with Entangled Photons, *Phys. Rev. Lett.* **115**, 250401 (2015).

[5] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, Proposed Experiment to Test Local Hidden-Variable Theories, *Phys. Rev. Lett.* **23**, 880 (1969).

[6] R. Penrose, Singularities and Time-Asymmetry, in *General Relativity: An Einstein Centenary Survey*, Cambridge University Press, pp. 581–638 (1979).

[7] P. R. Holland, *The Quantum Theory of Motion*, Cambridge University Press (1993).

[8] G. Perelman, The Entropy Formula for the Ricci Flow and its Geometric Applications, *arXiv:math/0211159* (2002).

[9] G. Perelman, Ricci Flow with Surgery on Three-Manifolds, *arXiv:math/0303109* (2003).

[10] G. Perelman, Finite Extinction Time for the Solutions to the Ricci Flow on Certain Three-Manifolds, *arXiv:math/0307245* (2003).

[11] R. S. Hamilton, Three-manifolds with positive Ricci curvature, *J. Differential Geom.* **17**(2), 255–306 (1982).

[12] J. B. Hartle and S. W. Hawking, Wave Function of the Universe, *Phys. Rev. D* **28**, 2960 (1983).

[13] P. A. M. Dirac, Quantised Singularities in the Electromagnetic Field, *Proc. R. Soc. London A* **133**, 60–72 (1931).

[14] E. Zeckendorf, Représentation des nombres naturels par une somme de nombres de Fibonacci, *Bull. Soc. R. Sci. Liège* **41**, 179–182 (1972).

[15] B. S. Tsirelson, Quantum Generalizations of Bell's Inequality, *Lett. Math. Phys.* **4**, 93–100 (1980).

[16] B. S. DeWitt, Quantum Theory of Gravity. I. The Canonical Theory, *Phys. Rev.* **160**, 1113 (1967).

[17] J. A. Wheeler, Superspace and the Nature of Quantum Geometrodynamics, in *Batelle Rencontres*, Benjamin, pp. 242–307 (1968).

[18] A. Preece and B. Batenin, Canonical Quantization of GR on $\mathbb{R} \times S^3$ in CMC Gauge and a Phase Law for Cosmic Acceleration, Zenodo *preprint* (2025).

[19] N. Aronszajn, A unique continuation theorem for solutions of elliptic partial differential equations or inequalities of second order, *J. Math. Pures Appl.* **36**, 235–249 (1957).

[20] S. Helgason, *Groups and Geometric Analysis*, American Mathematical Society (2000). [Paley–Wiener theorem for Riemannian symmetric spaces.]

[21] M. V. Berry, Quantal phase factors accompanying adiabatic changes, *Proc. R. Soc. A* **392**, 45–57 (1984).

[22] V. Coffman, J. Kundu, and W. K. Wootters, Distributed entanglement, *Phys. Rev. A* **61**, 052306 (2000).

[23] CMB-S4 Collaboration, CMB-S4 Science Book, First Edition, *arXiv:1610.02743* (2016).

[24] LiteBIRD Collaboration, Probing Cosmic Inflation with the LiteBIRD CMB Polarization Survey, *PTEP* **2023**, 042F01 (2023).

[25] N. Cornish, D. Spergel, and G. Starkman, Circles in the sky: finding topology with the microwave background radiation, *Class. Quantum Grav.* **15**, 2657 (1998).

[26] Planck Collaboration, Planck 2015 results. XVIII. Background geometry and topology, *A&A* **594**, A18 (2016).

[27] DESI Collaboration, DESI 2025 results II: Baryon Acoustic Oscillations, *Phys. Rev. D* **112**, 083515 (2025).

[28] DESI Collaboration, Extended dark energy analysis using DESI DR2 BAO, *Phys. Rev. D* **112**, 083511 (2025).

[29] J. Preskill, Lecture Notes on Topological Quantum Computation, Chapter 9, `http://www.theory.caltech.edu/~preskill/ph219/` (2004).

[30] A. Preece and B. Batenin, Wave Function of the Void: Dual-Sector Quantum Cosmology on S^3, Zenodo *preprint* (2026).